

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch :Mechanical & Allied Engineering

Semester :V

Subject Code & Name: Applied Thermodynamics (BTMC506)

Max Marks: 60

Date:12/01/2024

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*
- 5. Use of steam table is allowed.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain the Orsat apparatus for flue gas analysis with neat schematic diagram.	CO1	6
B) Develop the equation for the measurement of calorific value of solid fuel by the use of bomb calorimeter with neat sketch.	CO1	6
C) The flue gas analysis by volume for burning of C_6H_{14} is as follows: $CO_2 = 8.5\%$, $CO = 7.8\%$, and $N = 83.7\%$ Calculate (a) the air-fuel ratio required for chemically correct combustion.	CO1	6
Q.2 Solve Any Two of the following.		12
A) A boiler generates 8.5 kg of steam per kg of coal burned at a pressure of 13.5 bar from feed water having absolute temperature of 350 K. The boiler efficiency is 70% and factor of evaporation 1.17. Taking specific heat of steam at constant pressure to be 2.1 kJ/kg K, calculate (a) the degree of superheat and temperature of steam generated, (b) the calorific value of coal, and (c) the equivalent evaporation.	CO1	6
B) Define and explain mountings and accessories with suitable any one schematic diagram of both.	CO3	6
C) Derive the expression for the height of natural draught.	CO3	6
Q. 3 Solve Any Two of the following.		12
A) Steam expands from 2.5 bar to 1 bar in a nozzle. The initial velocity of steam is 80 m/s and initial temperature is 200°C. Taking nozzle efficiency as 96%, find the exit velocity.	CO2	6
B) Derive the equation for thermal efficiency of Reheat Rankine cycle with neat sketch.	CO2	6
C) Derive and explain the thermal efficiency of Joule Bryton cycle.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Explain the velocity and pressure compounding of impulse turbine with suitable schematic diagram.	CO4	6

B)	In a single stage impulse turbine, the blade angles are equal and the nozzle angle is 20° . The velocity coefficient for the blade is 0.83. Find the maximum blade efficiency possible. If the actual blade efficiency is 90% of maximum blade efficiency, find the possible ratio of blade speed to steam speed.	CO4	6
C)	Differentiate the Jet and surface condenser with neat sketch.	CO4	6
Q. 5 Solve Any Two of the following.			12
A)	Explain reciprocating air compressor with neat sketch.	CO5	6
B)	Explain the benefits of the multi staging in reciprocating air compressor with PV diagram.	CO5	6
C)	Derive the equation for indicated work of single stage, single acting reciprocating air compressor with neat sketch.	CO5	6

*** End ***